

KrishiGo – Smart Agricultural Equipment Rental Platform

A Comprehensive Conceptual, Technical, and Business Study

1. Abstract

Agriculture remains the backbone of India's economy, yet small and marginal farmers struggle with limited access to modern equipment due to high ownership costs and irregular income cycles. **KrishiGo** aims to solve this challenge by creating a **digital marketplace** that connects farmers who need machinery with equipment owners willing to rent them out.

By leveraging mobile technology, geolocation, and digital payments, KrishiGo enables affordable, efficient, and transparent farm equipment access—improving productivity, reducing idle asset time, and promoting sustainable agriculture.

2. Background and Motivation

In India, more than **85% of farmers are smallholders** with landholdings below 2 hectares.

High-cost machinery like tractors, harvesters, and rotavators often remain out of reach. Meanwhile, many large farmers or agri-businesses own such equipment that stays **idle for 60–70% of the year**.

This imbalance highlights an untapped opportunity: creating an organized rental ecosystem for agricultural machinery — similar to what **Uber did for transport** or **Zerodha did for trading**.

KrishiGo bridges this gap by:

- Offering a **digital bridge** between equipment owners and farmers.

- Ensuring **on-demand access** to machinery without ownership.

- Building **trust and transparency** using verified profiles, reviews, and secure payments.

3. Problem Statement

Despite the growth of India's agri-tech sector, small farmers face persistent issues:

- **High Equipment Costs:** Machinery like tractors or seeders cost ₹5-15 lakh — unaffordable for smallholders.
 - **Low Utilization of Owned Equipment:** Owners recover costs slowly due to limited rental networks.
 - **Lack of Transparency:** Local middlemen dominate rentals with irregular pricing.
 - **Limited Awareness:** Farmers often don't know what equipment is available nearby.
 - **No Standardized Platform:** Most villages lack digital systems for booking or verifying equipment.
-

4. Objective of KrishiGo

KrishiGo's main objectives are:

- To **create a unified online platform** for agricultural equipment rentals.
 - To **empower small farmers** with affordable access to modern machinery.
 - To **maximize equipment utilization** for owners through wider reach.
 - To ensure **transparent pricing and reliable service** via digital verification and reviews.
 - To **collect agricultural data insights** for predictive analytics and optimization.
-

5. Scope of the Project

Primary Users:

- ◆ Farmers seeking equipment (renters).
- ◆ Equipment owners (suppliers).
- ◆ Local service providers (mechanics, transporters).

Geographical Focus:

Tier 2, Tier 3 towns and rural regions across India.

Technology Scope:

Mobile-first platform (Android app + web dashboard).

Location-based search and booking.

Digital wallet & UPI integration for payments.

Data analytics for utilization and demand forecasting.

6. Proposed Solution – KrishiGo Overview

KrishiGo is a **two-sided digital marketplace** for agriculture equipment rentals.

It functions like “Ola for tractors” or “Zerodha for agri-assets”, bringing simplicity, transparency, and digital reliability to rural India.

How KrishiGo Works

- Equipment Listing:**
Owners upload their machines with details – type, price/hour, condition, location, photos.
- Farmer Search & Booking:**
Farmers enter their crop or task type (e.g., tilling, spraying), and KrishiGo suggests nearby equipment.
- Verification & Confirmation:**
The system verifies both parties (via KYC) and confirms rental availability.
- Usage & Payment:**
Once service is completed, payments are processed digitally with KrishiGo taking a small commission.
- Feedback & Ratings:**
Both farmers and owners rate the transaction, improving trust.

7. Key Features

Category	Features
Farmer Side	Search equipment, Book online, Pay via UPI, Give ratings
Owner Side	Add equipment, Manage bookings, Set prices, View analytics
Admin Side	Verify users, Manage disputes, Monitor platform data
Add-ons	AI-based equipment suggestions, repair service tie-ups

8. Vision and Mission

Vision:

To make modern agricultural tools accessible to every farmer in India, promoting efficiency and sustainability through shared resources.

Mission:

To build India's largest, most trusted agricultural equipment rental network powered by technology, data, and community trust.

9. System Architecture Overview

KrishiGo follows a **three-tier architecture**, ensuring scalability and performance.

1 Presentation Layer (Frontend)

Farmer & Owner mobile/web interfaces

Built using **React / Flutter**

Integrated with **Google Maps API**

2 Application Layer (Backend)

Business logic, user verification, booking system, payments

Technologies: **Node.js / Express.js** or **Django (Python)**

Integrations: Razorpay, Firebase, Twilio

3 Database Layer

Stores all persistent data — users, equipment, transactions, reviews

Database: **MongoDB / PostgreSQL**

10. Workflow Diagram (Text Format)

Farmer → Search → KrishiGo App → Backend → Equipment
OwnerOwner → Confirm → Payment Gateway → Database →
Notification to both

11. Design Details

Frontend

Regional language support

Tabs: Home, Bookings, Add Equipment, Profile

Tech: React, TailwindCSS, Material UI

Backend

RESTful APIs

JWT authentication

Data Models:

User(id, role, name, location)

Equipment(id, owner_id, type, cost, status)

Booking(id, farmer_id, equipment_id, total_cost)

Review(id, rating, comment)

12. Core Algorithms

Equipment Matching

```
function findEquipment(taskType, location):  
    allEquipment = fetchFromDatabase()  
    filtered = []  
    for eq in allEquipment:  
        if eq.type == taskType and eq.available:  
            eq.distance = calcDistance(location, eq.location)  
            filtered.append(eq)  
    return sort(filtered, by=['distance','rating','price'])
```

Dynamic Pricing

$$\text{Rental_Price} = \text{Base_Price} \times (1 + \text{Demand_Factor} - \text{Condition_Factor})$$

★ **Trust & Rating**

$$\text{Weighted_Rating} = (\text{Recent_Ratings} \times 0.7) + (\text{Older_Ratings} \times 0.3)$$

13. Methodology

Phase	Description	Output
1	Research farmer pain points	Requirement Report
2	Prototype design	UI mockups
3	Backend setup	Database + APIs
4	Frontend integration	Functional prototype
5	Testing	Beta release
6	Deployment	Live version
7	Scaling	New features

14. Revenue Model

Commission Model

5-10% fee per transaction.

Example: ₹200 commission on ₹2000 rental.

Subscription Model

Tier	Price/Month	Benefits
Basic	Free	Standard listing
Silver	₹499	Verified badge
Gold	₹999	Analytics, ads, promotion

Add-on Services

Maintenance and repair

Equipment insurance

Agri-lending & fuel delivery

Data & Advertisement

Ad space for agri brands

Rental data insights for partners

15. Financial Projection

Year	Users	Rentals/Month	Avg Commission	Monthly Revenue
1	10,000	5,000	₹200	₹10,00,000
2	50,000	25,000	₹250	₹62,50,000
3	2,00,000	1,00,000	₹300	₹3,00,00,000

16. Social & Economic Impact

Economic

Reduces machinery costs.

Improves asset utilization.

Creates new rural income streams.

Social

Encourages collaboration among farmers.

Promotes rural digital literacy.

Eliminates middlemen exploitation.

Environmental

Shared usage reduces carbon footprint.

Promotes sustainable resource management.

17. Risk & Mitigation

Risk	Description	Mitigation
Low Adoption	Farmers hesitant to use apps	Local language UI, training camps
Payment Issues	Rental defaults	Escrow & UPI verification
Trust Issues	Fear of fraud	Ratings, KYC, verified badges
Seasonal Demand	Low off-season usage	Offer repair/maintenance jobs

18. Future Scope

AI Predictive Insights – Suggest optimal pricing and rental demand.

IoT Integration – GPS tracking, usage analytics for machines.

Voice Assistant – Multilingual booking support.

Financial Partnerships – Loans and insurance integration.

Government Tie-ups – Integration with AgriTech missions.

19. Workflow Summary

Farmer → Login → Search → Pay → Use → Rate Owner → Add → Accept → Get Paid → Rate Admin → Manage → Monitor → Analyze Data

20. Technology Stack

Category	Tools
Frontend	React / Flutter
Backend	Node.js / Django
Database	MongoDB / PostgreSQL
Hosting	AWS / Firebase
Payments	Razorpay / Paytm
Maps	Google Maps API

Category	Tools
Analytics	PowerBI / Metabase
Notifications	Firebase / Twilio

21. References

Government of India, Agriculture Census 2021

Plantix (PEAT GmbH) Case Study

FAO Mechanization Report 2023

Zerodha Startup Model Analysis

Indian AgriTech Market Report 2024 – NASSCOM

McKinsey Insights – Digital Agriculture in India

22. Conclusion

KrishiGo bridges the rural technology gap by turning expensive agricultural machinery into shared, affordable assets. It combines **technology, community, and sustainability** – helping farmers earn more, waste less, and farm smarter.

By adopting a simple yet scalable digital model, KrishiGo can become the **“Uber for farm equipment”**, revolutionizing India’s agricultural landscape and empowering millions of farmers to thrive in a modern, connected economy.